JULIAN ESTEBAN RINCÓN RODRÍGUEZ

Computer Science & Artificial Intelligence

Bogotá, Colombia • julianer2002@gmail.com • +57 311 828 3342 • github.com/Julian-Rincon

PROFESSIONAL PROFILE

8th-semester Computer Science & AI student at Universidad Sergio Arboleda, specialized in data science, machine learning, and intelligent systems. Hands-on experience building data pipelines, training predictive models (XGBoost, LightGBM, neural networks), cloud infrastructure on AWS, and local AI applications. Proven ability to take projects beyond the academic setting and deliver production-grade, documented work.

EDUCATION

Universidad Sergio Arboleda — Bogotá, Colombia *2021 – Present*

B.Sc. Computer Science & Artificial Intelligence | 9th semester of 9

GPA: 4.04/5.0 (Colombian scale) · German equivalent ≈2.4 (Modified Bavarian Formula) · Upward trend: last 3 semesters avg. 4.5+/5.0

Current courses (8th sem.): Big Data & Data Engineering, Machine Learning, Simulation & Modeling, Neural Networks — Classification (Elective), Engineering Communication, Ethics & Social Responsibility.

FEATURED PROJECTS

NEXUS — Autonomous Personal AI System *2025 – Present*

Python 3.11 · 5-tier LLM routing (Cerebras/NVIDIA NIM) · GCP hybrid deploy · Custom wake word · Cloned voice (ElevenLabs) · MCP server · ChromaDB

- Architected 5-tier LLM routing (Ollama → Groq → Cerebras 2600 tok/s → NVIDIA NIM 128K-context multimodal → Gemini fallback) with 38 auto-discovered skills spanning home automation, personal finance parsing, and passive security monitoring.
- Built an MCP server exposing system capabilities as tools for external AI agents; deployed a hybrid cloud/local architecture on a free-tier GCP VM (Tailscale-tunneled, service-account impersonation) running core services 24/7 with automated cost-guard cutoff.
- Designed safety-first automation: confirmation-gated system power control and allowlisted actions ensure irreversible commands never fire on a single ambiguous phrase; custom-trained wake word (ONNX, no PyTorch) and cloned voice via ElevenLabs with local fallback.
- Designed safety-first automation: confirmation-gated system power control, allowlisted actions, and passive-only security monitoring; custom-trained wake word and cloned-voice TTS with local fallback.

SAVI v2 — Autonomous Real Estate Valuation System *2026*

Python · XGBoost · K-Means · MDP · Q-Learning · Deep Q-Network · PyTorch · scikit-learn

- Architected full RL decision pipeline: MDP Value Iteration (convergence 259 iter, γ=0.95) → Q-Learning tabular (8K episodes, ε-greedy decay) → Double DQN (PyTorch, 150 epochs, replay buffer 20K, Adam lr=3e-4).
- Consensus policy across all three agents autonomously APPROVE/REVIEW/REJECTs valuations based on expected financial risk; XGBoost + cluster feature achieves R^2=0.9609 on 20,203 properties (2006–2024).
- Delivered IEEE-format technical paper, interactive HTML demo on GitHub Pages, and full reproducible Python pipeline — published to production portfolio.

Project Dogma — Stochastic Social Propagation Engine 2026

TypeScript · React · Dynamic Markov Chains · Hidden Markov Model · Nash Equilibrium · Monte Carlo

- Architected a mathematical simulation engine for ideological propagation across urban social graphs: Dynamic Markov Chains model agent state transitions (Neutral → Believer → Fanatic) with per-agent solitude, skepticism, and charisma variables modifying transition matrices each tick.
- Modeled authority behavior via Hidden Markov Model (Forward Algorithm): four latent states inferred from a single observable signal (Heat entropy), generating probabilistic threat cycles without exposing internal state.

- Resolved doctrinal conflicts via 2ൠ Nash Equilibrium mixed strategy (55%/45%), eliminating dominant pure strategies; resource economy governed by Verhulst logistic differential equation (carrying capacity K). Validated across 500 Monte Carlo simulations: μ=152 ticks, conversion 5.42 agents/tick (σ=0.39), retention ratio 38:1.

ShopStream — End-to-End AWS Big Data Pipeline *2026*

Python · PySpark · AWS Lambda · AWS EMR · AWS Glue · RDS PostgreSQL · Flask · Zappa · GitHub Actions

- Engineered production-grade data pipeline processing 2.5M synthetic behavioral events (page_view, click, search, product_view, cart_event) partitioned by date across S3 raw and processed zones.
- Deployed S3-triggered Lambda validator with CloudWatch custom metrics and automatic quarantine of invalid records; EMR/PySpark job computes 6 business KPIs including bounce rate, conversion funnel, and anomaly detection.
- Delivered Flask REST API on AWS Lambda via Zappa exposing DWH analytics endpoints; 86% test coverage with pytest; CI/CD via GitHub Actions (lint, test, deploy, smoke-test); RDS security group hardened from 0.0.0.0/0 to /32 CIDR.

Project Dogma — Stochastic Social Propagation Engine *2026*

TypeScript · React · Dynamic Markov Chains · HMM · Verhulst ODE · Nash Equilibrium · Monte Carlo

- Architected stochastic social propagation simulator modeling ideological spread across urban agent graphs; Dynamic Markov Chains govern state transitions (Neutral → Believer → Fanatic) modulated by per-agent solitude, skepticism, and charisma variables.
- Implemented HMM (Forward algorithm) for authority behavior inference from latent states via observable Heat entropy signal; Verhulst logistic differential equation governs resource carrying capacity K, generating S-curve propagation dynamics.
- Resolved doctrinal conflict via 2×2 Nash Equilibrium mixed strategy (55%/45%); validated across 500 Monte Carlo simulations: μ=152 ticks to convergence, conversion 5.42 agents/tick (σ=0.39), retention ratio 38:1.

Big Data Platform — AWS (Chinook / Sakila / Credit) *2026*

AWS EC2 · RDS (PostgreSQL / MySQL) · Athena · S3 · Glue · Terraform · React · FastAPI

- Fullstack on Chinook DB: React/Vite frontend + FastAPI backend on separate EC2 instances, PostgreSQL on RDS; 17-section technical document.
- Hive-partitioned external table (credit2) on Athena + S3; resolved case-sensitivity issue in partitions with manual ALTER TABLE.
- NYC Yellow Taxi dataset (94M rows, 12 months) on AWS EMR with PySpark; resolved schema inconsistency across Parquet files using explicit StructType and per-file normalization with unionByName.

Neural Networks & RL Labs *2026*

PyTorch · TensorFlow · CUDA · CIFAR-10 · OpenAI Gym

- Reinforcement learning: Q-learning (FrozenLake), REINFORCE (CartPole), and DQN with replay buffer.
- CNN labs: manual convolution, SmallCNN with dropout and weight decay trained on CIFAR-10; professional Word reports.

Ames Housing ML Pipeline *2026*

Python · XGBoost · LightGBM · scikit-learn · Pandas · Jupyter Notebook

- Best model: LightGBM R^2 = 0.75, MAE ≈ $23,600 on ~20,000 records (2006–2024 dataset).
- Complete pipeline: EDA, feature engineering, K-Means and hierarchical clustering, comparison of 6 algorithms.

Custom Programming Language *2023*

Python · ANTLR4 · Tokenization · Interpretation

- Designed and implemented a Python-inspired language with custom tokenizer, lexer, parser, and visitor-pattern execution engine using ANTLR4.

TECHNICAL SKILLS

Languages	Python (intermediate–advanced), Bash Scripting, JavaScript (basic), SQL
ML / AI	scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow, Whisper, Ollama
Big Data & Cloud	AWS (EC2, RDS, S3, Athena, Glue, EMR), Terraform, Git / GitHub, Docker (basic)
Data & ETL	Pandas, NumPy, Matplotlib, openpyxl, Power BI, Jupyter Notebook, PySpark
Automation	Playwright, pywinauto, Telegram Bot API, Gmail / Calendar OAuth2
Other	ANTLR4, CUDA, NetworkX, ChromaDB, ZeroMQ, tkinter, pytest

ACHIEVEMENTS & RECOGNITION

🏆 **International Award Winner — WarmiTics (Wisibilízalas, Universidad Pompeu Fabra)** *2019*

Universidad Pompeu Fabra · Barcelona, Spain · Shakespeare School, Quito, Ecuador · Senior Category

- Served as Technical Product Manager and Lead Frontend/UX Developer for WarmiTics, a web platform highlighting women leaders in STEM. 1st place, Senior category, age 17, representing Shakespeare School (Ecuador) against teams from Latin America and Europe.
- Architected and developed the full web application end-to-end; managed cross-functional logistics and stakeholder coordination; conducted structured interviews with women leaders in technology to gather primary content.
- Project available at: sites.google.com/view/warmitics

Independent Project — NEXUS

Self-directed development outside the academic curriculum

- Autonomous personal AI system built from scratch — manages the machine, not just conversation. 38 skills, MCP server, hybrid cloud/local deploy (GCP), safety-gated automation. Personal initiative beyond the academic curriculum.

CERTIFICATIONS

Database Design — Oracle Academy — August 2024

Database Programming with SQL — Oracle Academy — December 2024

AWS Academy Graduate — Data Engineering — Amazon Web Services Academy — May 2026 — 40 hours · Credly: credly.com/go/UGp1CUEB

LANGUAGES

Spanish: Native • English: B2+ (technical and academic use — documentation, presentations, reports)

Available for internships, freelance projects, or collaborations in AI, Data Science, and Software Development.